Data Description

Important notes:

- All of the data is synthetic. Resemblance to real people or situations is incidental.
- Use only data from this mini challenge to answer the questions. Data may be inconsistent between challenges. No external sources are required to answer the questions posed.

Graph Description

- File name: data.json
- Directed Multigraph
- # nodes: 17,412
- # edges: 37, 857
- 18 connected components
- Possible node types are: {Person, Song, RecordLabel, Album, MusicalGroup}
- Possible edge types are: {MemberOf, PerformerOf, ComposerOf, ProducerOf, LyricistOf, InStyleOf, InterpolatesFrom, CoverOf, LyricalReferenceTo, DirectlySamples, RecordedBy, DistributedBy}
- The graph format is a JSON file generated by Python's network.node_link_data() function. It can likewise be loaded to a networkx object using the corresponding node_link_graph() function. The root-level JSON object consists of graph-level properties specifying that it is directed and a multigraph, a "nodes" key which holds the list of nodes, and a "links" key which holds the list of edges.

The data for this challenge comes from two different sources

- an online, crowdsourced repository of musical influence, where contributors have manually notated instances in which songs or albums have sampled, covered, or otherwise drawn inspiration from previous work.
- aggregate-level descriptions of song popularity, as labeled by our journalist, Silas Reed. These notations (*notable* and *notoriety_date*) are directly based on both the number of sales and streams that the song/album has achieved as well as the whether or not the work landed on a top record chart.

Graph Schema

Nodes

Person nodes: These can be anyone in the music industry, including singers, producers, instrumentalists, composers, etc.

- **Node Type (string)** – the type of node
- **name (string)** – the name of the person
- **stage_name (string)** – if provided, the stage name of the musician

MusicalGroup nodes: Bands, quartets, small choirs, or other officially organized entities formed by musicians to make music.

- **Node Type (string)** – the type of node
- **name (string)** – the name of the person

RecordLabel nodes: These are organizations—professional, commercial, or otherwise institutional—involved in the recording, production, or distribution of the music.

- **Node Type (string)** – the type of node
- **Name (string)** – the name of the organization

Song nodes:

- **Node Type (string)** – the type of node
- **single (boolean)** – if provided, specifies whether or not the song was released as a standalone, or as part of a larger album
- **genre (string)** – the song's genre
- **notable (boolean)** – whether or not the song has appeared on a top record chart
- **release_date (string)** – the year in which the song was released
- **notoriety_date (string)** – if provided, the year in which the song first appeared on a top record chart
- **written_date (string)** – if provided, the year in which the song was written

Album nodes:

- **Node Type (string)** – the type of node
- **genre (string)** – the album's genre
- **notable (boolean)** – whether or not the album has appeared on a top record chart
- **release_date (string)** – the year in which the album was released
- **notoriety_date (string)** – if provided, the year in which the album first appeared on a top record chart

- **written_date (string)** – if provided, the year in which the album was written

Edges

For all of the edges in this graph, the only property is **Edge Type**. These types are listed below.

- **PerformerOf** – Indicates a that the source node (Person or MusicalGroup) performed the destination node (Song or Album)
- **ComposerOf** – Indicates that the source node (Person) composed the destination node (Song or Album)
- **ProducerOf** – Indicates that the source node (Person or RecordLabel) participated in the production of the destination node's work (Song, Album, Person, or MusicalGroup)
- **LyricistOf** – Indicates that the source node (Person) wrote lyrics for the destination node (Song or Album)
- **RecordedBy** – Indicates that the destination node (RecordLabel) aided in the recording process for the source node (Song or Album)
- **DistributedBy –** Indicates that the destination node (RecordLabel) aided in the distribution process for the source node (Song or Album)
- **InStyleOf –** Indicates that the source node (Song or Album) was performed at least partly in the style of the destination node (Song, Album, Person, or MusicalGroup)
- **InterpolatesFrom –** Indicates that the source node (Song or Album) interpolated a melody from the destination node (Song or Album).
- **CoverOf –** Indicates that the source node (Song or Album) is a cover of the destination node (Song or Album)
- **LyricalReferenceTo –** Indicates that the source node (Song or Album) makes a lyrical reference to the destination node (Song or Album)
- **DirectlySamples –** Indicates that the source node (Song or Album) consists of (an) audio recording(s) that directly reuse a portion of the audio recording of the destination node (Song or Album) via sampling
- **MemberOf** – Indicates that the source node (Person) is (or was) a member of the destination node (MusicalGroup)